

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 int compare(const void *a, const void *b) {
4     long long x = *(const long long *)a;
5     long long y = *(const long long *)b;
6     if (x < y) return -1;
7     if (x > y) return 1;
8     return 0;
9 }
10 int main() {
11     int n;
12     if (scanf("%d", &n) != 1) return 0;
13     long long arr[n];
14     for (int i = 0; i < n; i++) {
15         scanf("%lld", &arr[i]);
16     }
17     qsort(arr, n, sizeof(long long), compare);
18
19     long long max_sum = 0;
20     for (int i = 0; i < n; i++) {
21         max_sum += arr[i] * i;
22     }
23     printf("%lld\n", max_sum);
24
25     return 0;
26 }
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct